ASCII MASTER FLOW — PANEL 1/12: Weather causation chain

```
INSOLATION -> surface and atmospheric heating
TEMPERATURE GRADIENT -> pressure gradient
WIND -> moisture transport and convergence
LIFT + SATURATION -> cloud and precipitation
MUST REMEMBER: Weather elements connect radiation balance, temperature, pressure-gradient
  force, Coriolis/friction, humidity, stability, condensation, clouds and precipitation;
  upper-air Rossby waves, subtropical jet and western-disturbance tracks anchor Indian
  seasonality.
```

ASCII MASTER FLOW — PANEL 2/12: Heat-budget ledger

```
SUN -> incoming shortwave
SURFACE -> reflected + absorbed energy
EARTH -> longwave + sensible + latent heat
BALANCE -> temperature and circulation response
```

ASCII MASTER FLOW — PANEL 3/12: Stability profiles

```
NORMAL TENDENCY -> temperature falls upward
INVERSION -> temperature rises through layer
STABLE LAYER -> vertical mixing suppressed
SURFACE INVERSION -> fog and pollution trapping
```

ASCII MASTER FLOW — PANEL 4/12: Wind-force triangle

```
PRESSURE GRADIENT -> starts acceleration
CORIOLIS -> right north / left south
FRICTION -> slows and crosses isobars
ALOFT -> near-geostrophic parallel flow
```

ASCII MASTER FLOW — PANEL 5/12: Planetary circulation rail

```
EQUATORIAL LOW -> convergence and ascent
SUBTROPICAL HIGH -> descent and trades
SUBPOLAR LOW -> westerly-polar interaction
POLAR HIGH -> cold outflow; belts migrate
```

ASCII MASTER FLOW — PANEL 6/12: Humidity-to-cloud chain

```
MOISTURE CONTENT + TEMPERATURE -> relative humidity
COOLING TO SATURATION -> dew point reached
NUCLEI + LIFT -> droplets or ice crystals
GROWTH + FALL SPEED -> precipitation
```

ASCII MASTER FLOW — PANEL 7/12: Rainfall comparison

```
CONVECTIONAL -> buoyant thermal ascent
OROGRAPHIC -> forced relief ascent
FRONTAL / CYCLONIC -> convergent dynamic ascent
REAL EVENT -> mechanisms may combine
```

ASCII MASTER FLOW — PANEL 8/12: Troposphere section

```
SURFACE -> moisture, friction and heating
MID LEVELS -> clouds, troughs and stability
UPPER TROPOSPHERE -> jets and divergence
TROPICS DEEPER / POLES SHALLOWER
```

ASCII MASTER FLOW — PANEL 9/12: Jet-stream engine

```
HORIZONTAL TEMPERATURE GRADIENT
HEIGHT / PRESSURE GRADIENT ALOFT
CORIOLIS BALANCE -> strong near-zonal wind
ROSSBY WAVES -> ridges, troughs and steering
```

ASCII MASTER FLOW — PANEL 10/12: India seasonal jets

```
WINTER -> STWJ south of Himalaya sector
STWJ -> steers western disturbances east
SUMMER -> tropical easterly jet develops
RULE -> jets are parts of wider circulation
CLOSE DISTINCTION: Humidity, relative humidity and dew point differ; weather differs from
  climate, jet stream from surface wind, and a western disturbance is an eastward-moving
  extratropical system embedded in westerlies rather than a western Indian monsoon branch.
```

ASCII MASTER FLOW — PANEL 11/12: Western-disturbance chain

```
UPPER TROUGH / LOW -> eastward westerly motion
MOISTURE + DYNAMICAL LIFT -> cloud and rain/snow
HIMALAYA + PLAINS -> terrain-specific impacts
OUTCOME -> beneficial or damaging by timing/intensity
EVIDENCE LIMIT: Jet position is one control among topography, blocking, moisture supply and
  synoptic evolution; event attribution and seasonal forecasts require dated agency evidence
  and probabilistic language.
```

ASCII MASTER FLOW — PANEL 12/12: Forecast evidence ladder

```
RADIOSONDE / PILOT BALLOON -> upper-air state
SATELLITE + RADAR -> cloud and precipitation
NWP ENSEMBLE -> scenarios and uncertainty
WARNING -> action; later observations verify outcome
```